Transitive First: Gradient Descent Learns the Ranking Before the Cycles — a Hodge Decomposition of the Pairwise Neural Tangent Kernel

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Abstract

Pairwise preferences — which response is better, which player wins — decompose on the comparison graph into a transitive part (a global ranking), local cycles (rock–paper–scissors) and global cycles, the combinatorial Hodge decomposition behind HodgeRank. Scalar reward models of the Bradley–Terry type cannot represent cycles; skew-symmetric models can. We ask the dynamical question that neither literature addresses: when a neural preference model is trained by gradient descent, *in which order* does it learn these parts? We write the neural tangent kernel (NTK) between *edges* for three models — a scalar reward, an antisymmetric network $h(x,y) - h(y,x)$, and the General Preference Model — and project it on the Hodge subspaces. For the scalar model the kernel annihilates every cycle exactly (measured $\leq 3 \times 10^{-16}$). For the antisymmetric network, in all 96 graphs we examined the mean kernel eigenvalue on the gradient subspace exceeds that on cycle space by a factor of 4.0 to 123, growing with the mean degree of the graph; the top eigenvectors are 99% transitive. The consequence is a spectral bias: in 12 of 12 training runs, gradient descent fits the transitive part first and cycles 25× later (GPM: 12×), under the squared loss — where the linearised NTK prediction tracks training to within 0.02 — and under the Bradley–Terry loss on votes; the order survives feature learning (24/24 runs across width and lazy/rich scale) although the gap narrows. Two predictions fail and we report them: curl and harmonic cycles are learned at the same rate (ratio 0.80–1.71), and early stopping at the moment the ranking is fitted is too early, because it also discards real cycles. We prove that i.i.d. noise puts only a fraction $(|V|-1)/|E|$ of its energy in gradient space (measured within 0.0048 in 9 graphs), but show that annotator vote flips are different: they mostly shrink the transitive signal. Early stopping nonetheless buys up to 12.0 accuracy points for GPM at 40% flipped votes. On 52,200 Chatbot Arena votes among 63 models the same order holds on real data, and the 15.3% of cyclic energy in the votes is *less* than a perfectly transitive model with the same counts would produce (17.5%, $p = 0.96$): at the model level, Arena cycles are sampling noise, and a scalar reward is the best held-out predictor.

1 Introduction

Reward models for language-model alignment, rating systems for games, and recommenders trained on clicks all learn from the same kind of data: pairs. Response B was preferred to response A; player j beat player i. A set of such comparisons lives on the edges of a *comparison graph*, and combinatorial Hodge theory [7] splits every edge flow orthogonally into three parts (Figure 1): a *gradient* part, the differences of a global score — a ranking; a *curl* part, circulation around triangles — rock–paper–scissors; and a *harmonic* part, circulation around larger holes of the graph. A scalar reward model, the Bradley–Terry workhorse of RLHF [2, 12], can only express the first. Skew-symmetric models such as the General Preference Model (GPM) [16] and hybrid reward–cyclic decompositions [1] were proposed precisely to capture the other two.

That literature asks what a model *can* represent. We ask what gradient descent *learns first*. The question has

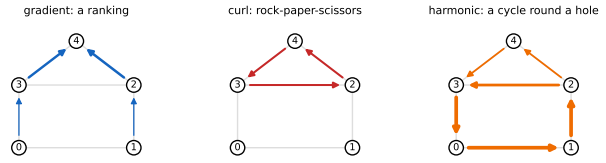


Figure 1: The three Hodge components of one edge flow on a five-item comparison graph (triangle 2–3–4 filled): a ranking, a local cycle, and a cycle around the square hole. Arrow width is flow magnitude.

teeth. If a flexible preference model fits the ranking long before it fits the cycles, then (i) the number of training steps is an implicit prior on transitivity; (ii) early stopping acts as a transitivity filter, discarding cyclic structure whether it is real or noise; and (iii) since random label noise on a well-connected graph falls almost entirely in cycle space, that filter should be most useful exactly when annotators disagree. The neural tangent kernel [6] is the natural tool: in the linearised regime [10], gradient

descent fits each eigen-direction of the kernel at a rate set by its eigenvalue, and the infinite-width limit is a Gaussian process [11, 9] — here a GP preference model [5] with a specific, computable kernel. What is missing is the kernel *between edges* and its relation to the Hodge subspaces.

We compute that kernel for three models, in closed form at infinite width and empirically at finite width, and test four hypotheses: that the scalar kernel is blind to cycles (H0); that the antisymmetric kernel is spectrally ordered, gradient first (H1); that training follows that order (H2); and that noise is cyclic and early stopping filters it (H3). We then test the whole picture on 52,200 human votes from Chatbot Arena (H4). H0–H2 hold cleanly; parts of H1 and H3 do not, and we report where and why.

2 Related work

Neural tangent kernels and spectral bias. The NTK [6] describes gradient descent on wide networks as kernel regression; wide networks evolve as their linearisation [10], and their priors are Gaussian processes [11, 9]. Directions with large kernel eigenvalues are learned first — the spectral or frequency bias [13]. Output scaling moves a network between this lazy regime and feature learning [4]. We use all of these, but on a kernel defined over *pairs* and decomposed by graph topology rather than by frequency.

Hodge theory for rankings. HodgeRank [7] fits the gradient component by least squares and reads the curl and harmonic residuals as measures of inconsistency; later work uses it for crowdsourced ranking and Bayesian intransitive models [15]. No neural network or training dynamics is involved.

Intransitive preference models. Bradley–Terry [2] underlies RLHF reward models [12]. GPM [16] embeds responses and scores pairs with a skew-symmetric bilinear form; HRC [1] decomposes preferences into an explicit transitive scalar and a cyclic vector part. These works establish expressivity and downstream gains; we study the learning dynamics that any such model is subject to.

NTK for pairwise feedback. Neural dueling bandits analyse Bradley–Terry preference learning in the NTK regime to obtain confidence bounds [14]; they do not decompose the kernel by Hodge structure. The closest work is the Topological NTK [8], which reports a "topological spectral bias" for message passing on edge signals of simplicial complexes; it concerns graph neural networks on a fixed complex, not preference models over item features, and does not address Bradley–Terry losses, annotator noise or early stopping.

3 Hodge decomposition of the edge NTK

Comparison graphs. Items $1..n$ with features $x_i \in \mathbb{R}^d$; a graph with oriented edges (i, j), $i < j$; an edge flow $y \in \mathbb{R}^E$, y_{ij} the preference of j over i (a log-odds). With the gradient $B : \mathbb{R}^V \to \mathbb{R}^E$, $(Bs)_{ij} = s_j - s_i$, and the curl $C : \mathbb{R}^E \to \mathbb{R}^T$ over triangles, the Hodge decomposition is $\mathbb{R}^E = \operatorname{im} B \oplus \operatorname{im} C^\top \oplus \ker(BB^\top + C^\top C)$, with orthogonal projectors $P_{\text{grad}}, P_{\text{curl}}, P_{\text{harm}}$ of ranks $|V| - 1$ (connected graph), the number of independent triangles, and the first Betti number.

Three models and their edge kernels. A preference model predicts g_{ij}. The *scalar* model $g_{ij} = f(x_j) - f(x_i)$; the *antisymmetric* model $g_{ij} = h([x_i, x_j]) - h([x_j, x_i])$; GPM $g_{ij} = e(x_i)^\top R\, e(x_j)$ with R block-skew. Each is an MLP in the NTK parameterisation. Under gradient descent with step η on $\frac{1}{2E}\sum(g-y)^2$ the training residual evolves, in the linearised regime, as $r_{t+1} = (I - \frac{\eta}{E}\Theta_E)\, r_t$, where Θ_E is the NTK between edges. For the scalar model $\Theta_E = B\,\Theta_V B^\top$ with Θ_V the ordinary NTK between items; for the antisymmetric model, with $z = [x_i, x_j]$ and $\tilde{z} = [x_j, x_i]$,

$$\Theta_E(e, e') = K(z, z') - K(z, \tilde{z}') - K(\tilde{z}, z') + K(\tilde{z}, \tilde{z}'), \quad (1)$$

K the closed-form ReLU NTK [6] on concatenated inputs. We define the *Hodge rate* of a component as the mean eigenvalue of Θ_E on its subspace, $\kappa_c = \operatorname{tr}(P_c \Theta_E P_c)/\dim_c$ — the expected Rayleigh quotient of a random flow in that component, hence its expected learning speed.

Proposition 1 (the scalar kernel is blind to cycles). For any cyclic flow $y_c \in \ker B^\top$, $\Theta_E\, y_c = B\Theta_V B^\top y_c = 0$. The cyclic residual of a scalar model therefore never changes under gradient descent on the squared loss, at any width and in any regime: it is not merely unrepresentable but invisible to the gradient. Numerically $|\kappa_{\text{cyc}}| \le 3 \times 10^{-16}$ across all graphs.

Why the antisymmetric kernel prefers rankings. Split the kernel on concatenated inputs into a part additive in its two halves and an interaction part. The additive part yields, after antisymmetrisation, a term of the form $B\,\tilde{K}\,B^\top$ — the scalar structure again — whose action on cycles vanishes by Proposition 1, while on gradient flows it is amplified by the number of edges that share each item. Cycles are therefore learned only through the interaction part. This predicts (i) $\kappa_{\text{grad}} \gg \kappa_{\text{cyc}}$, (ii) a ratio that grows with the mean degree of the comparison graph, and (iii) no reason for curl and harmonic cycles to differ.

Lemma 2 (i.i.d. noise is mostly cyclic). If $z \sim \mathcal{N}(0, \sigma^2 I_E)$, rotational invariance makes $\|P_{\text{grad}} z\|^2 / \|z\|^2 \sim \text{Beta}(\frac{|V|-1}{2}, \frac{|E|-|V|+1}{2})$, with mean

$(|V| - 1)/|E|$. On a complete graph that is $2/n$: almost all noise lives where a transitive-first learner looks last.

4 Spectrum and learning dynamics

The spectrum is ordered. Figure 2 confirms H1 for the gradient-versus-cycle split. In every one of the 96 graphs the antisymmetric kernel's gradient rate exceeds its cyclic rate, by 4.0× on the sparsest graphs to 123× on complete ones (mean 13.8× at density 0.15, 63× at density 1), and the ratio grows with the mean degree, as the additive-part argument predicts. Low feature dimension and shallow networks sharpen it (48.4× at $d = 2$ versus 26.2× at $d = 32$; 34.6× at depth 1 versus 17.0× at depth 2), since both enlarge the interaction part of the kernel. The top ten eigenvectors carry 99% of their mass in gradient space. At finite width the empirical kernel reproduces the analytic ratio already at width 128 (14.5 versus 15.0), and the kernels agree entrywise to 4.5% at width 2048. GPM is ordered too but less sharply (4.2–5.8×): its bilinear form has no additive part to amplify rankings.

Curl and harmonic are not ordered. The second half of H1 fails. The ratio of curl to harmonic rates ranges from 0.80 to 1.71 with mean 1.02: the network does not distinguish cycles around triangles from cycles around holes. This matches the argument above — both live only in the interaction part — and it is where our setting differs from the Topological NTK, whose message passing is built on the Hodge Laplacian itself [8].

Training follows the spectrum. On a synthetic task (40 items, $d = 8$, density 0.2; the true preference combines a smooth utility and a skew interaction, giving 59/30/11% gradient/curl/harmonic energy), Figure 3 shows H2. The antisymmetric network fits the ranking in a handful of steps and the cycles 25× (curl) and 29× (harmonic) later under the squared loss, and 19× and 22× later under the Bradley–Terry loss on votes; GPM shows the same order with a smaller gap (12× and 19×). The order held in 12 of 12 runs. Under the squared loss the linearised NTK prediction tracks the network's per-component fit to within 0.02 at every step — the dynamics *is* the spectrum. The scalar model fits the ranking just as fast and never moves on cycles (final curl fit $\leq 2 \times 10^{-08}$), which costs it on held-out pairs: 0.681 sign accuracy versus 0.882 (antisymmetric) and 0.840 (GPM), because the true preferences are partly cyclic.

Feature learning narrows the gap but keeps the order. Figure 4 moves the network out of the kernel regime. With $\alpha = 10$ the kernel barely moves (relative drift 0.029) and curl reaches 90% 66× after the ranking; with $\alpha = 0.1$ the kernel changes by a factor of 6.6 and the gap shrinks to 17×, because the evolving kernel becomes less ordered (final rate ratio 12 versus 17). The order held in all 24 of 24 runs across widths 32–1024. Feature learning here also generalises slightly better (held-out 0.892 versus 0.868): it learns the cycles sooner.

5 Noise and early stopping

Random noise is cyclic — annotator flips are not. Lemma 2 holds to within 0.0048 in all 9 graphs (Figure 5, left); on the densest graph (80 items, 3160 edges) only 0.025 of the noise is transitive. On the synthetic task the prediction is 0.234 and Gaussian noise puts 0.20 there. Vote flips behave differently: 0.60 of the resulting perturbation lies in gradient space. A flip rate ϵ pulls every win probability toward one half, so its main effect is to *shrink* the true flow — which is mostly transitive — rather than to add independent noise. The clean "noise is cyclic" story therefore applies to measurement noise, not to annotators who randomly disagree.

Early stopping helps, but not where the ranking is fitted. Our predicted stopping time $t^\star$, the step at which the linearised gradient component is 95% fitted, is too early: held-out accuracy there is 0.750 against 0.872 at the best step for the antisymmetric net, because stopping when the ranking is learned also discards the *true* cycles. Early stopping still earns its keep, and more so as noise grows (Figure 5, middle and right). The best step moves earlier with noise (from 1974 to 236 steps for the antisymmetric net, 1534 to 84 for GPM). At 40% flipped votes it recovers 4.3 accuracy points for the antisymmetric net (0.752 → 0.795) and 12.0 for GPM (0.656 → 0.776). The scalar model, which cannot overfit cycles at all, gains 4.1. The right stopping time lies *between* the gradient and cycle time scales, and where it falls depends on how much of the cyclic content is signal. The spectrum tells us the two time scales; it cannot by itself tell us that ratio.

6 Chatbot Arena

We take 52,200 human votes between 63 LLMs from the public Chatbot Arena release [3], keep the 640 model pairs with at least 20 votes, count ties as half a win, and hold out 20% of votes per pair (7,126 decisive held-out votes). Items have no features, so each network sees a one-hot model identity.

The cycles in Arena are sampling noise. The observed flow has 84.7% gradient and 15.3% cyclic energy (the graph has 576 triangle dimensions and only 2 harmonic ones). Is that intransitivity? We simulate votes from the best transitive fit with the *same* per-pair counts:

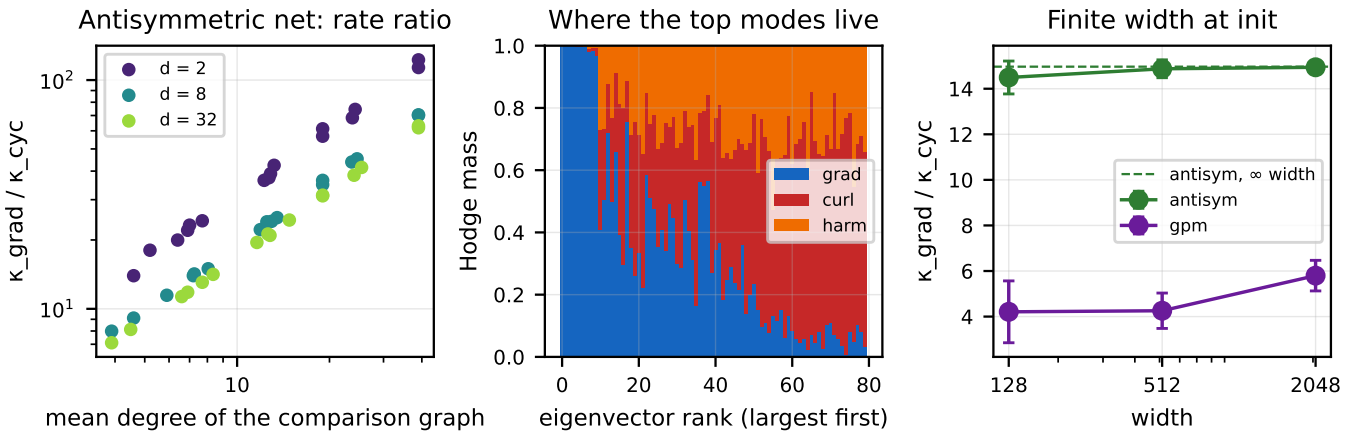


Figure 2: Edge NTK of the antisymmetric network. Left: ratio of the mean kernel eigenvalue on gradient flows to that on cyclic flows, over 96 random comparison graphs (sizes 20 and 40, densities 0.15–1, depth 1 shown), against the mean degree; colour is the feature dimension. Middle: Hodge composition of the top 80 eigenvectors on a 30-item graph (109 edges; gradient/curl/harmonic dimensions 29/51/29). Right: the same ratio for the empirical kernel at initialisation versus width, with the infinite-width value dashed; GPM is shown for comparison.

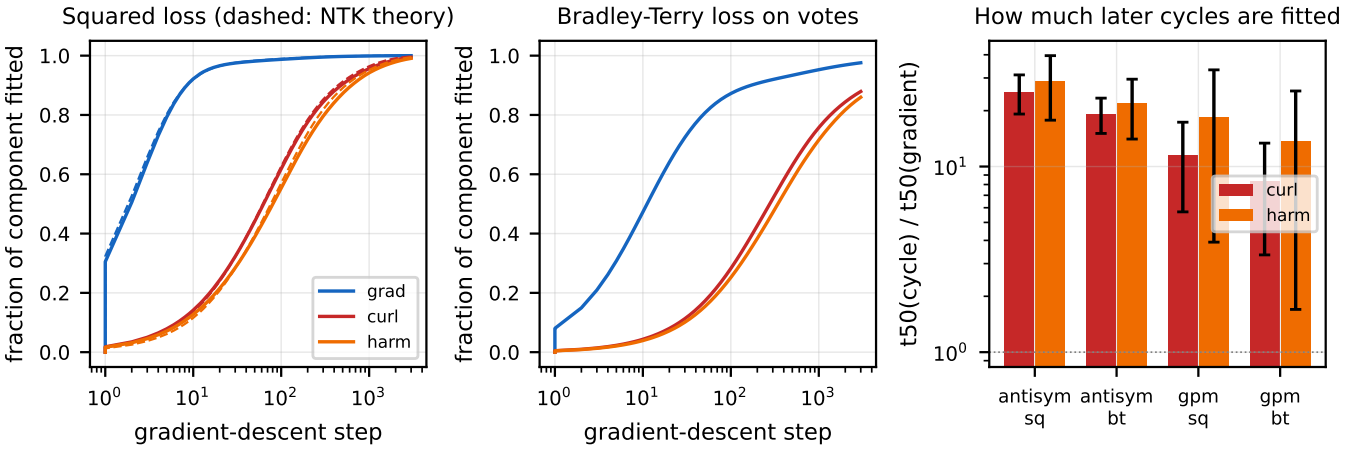


Figure 3: Fraction of each Hodge component of the true flow fitted during training (antisymmetric net, one seed). Left: squared loss, with the linearised NTK prediction dashed. Middle: Bradley–Terry loss on 200 votes per edge. Right: how many times later the cyclic components reach 50% than the gradient component, over seeds, for the antisymmetric net and GPM under both losses.

finite samples alone produce 17.5% cyclic energy (95% interval 15.2–19.7%). The observed value sits *below* the null ($p = 0.96$ for cycles at least as strong). At the level of models, Arena preferences show no intransitivity beyond what counting noise creates — which is Lemma 2 at work on real data.

The same order, and what it implies. On these votes the antisymmetric kernel ranks gradient over cycles by 44× (GPM 160×), and training fits the ranking by step 39 and the cycles only around step 1997 (Figure 6, middle). Since the cycles are noise, the flexible models' late phase is pure overfitting. Trained to the end, the scalar model is the best held-out predictor (log-loss 0.6282, accuracy 0.641, matching HodgeRank's 0.6284 and 0.641), ahead of the antisymmetric net (0.6317) and GPM (0.6306). With 40% of training votes flipped the gap widens (accuracy 0.622 versus 0.561 and 0.564), and early stopping recovers most of it (best 0.616 and 0.617). Stopped early, a cyclic model is a scalar model.

7 Limitations and conclusion

Limitations. The kernel analysis is exact at infinite width and for the squared loss; the Bradley–Terry and feature-learning results are empirical. Our networks are small MLPs on low-dimensional features, not reward models on top of a language model, where the "items" are responses with rich embeddings and each response appears in few comparisons — a sparse regime in which our ratio is smallest. The synthetic truth mixes one utility with one skew interaction; other cyclic structures could be learned at different speeds. The Arena analysis is at model level, so it says nothing about prompt-level intransitivity, where genuine cycles are more plausible,

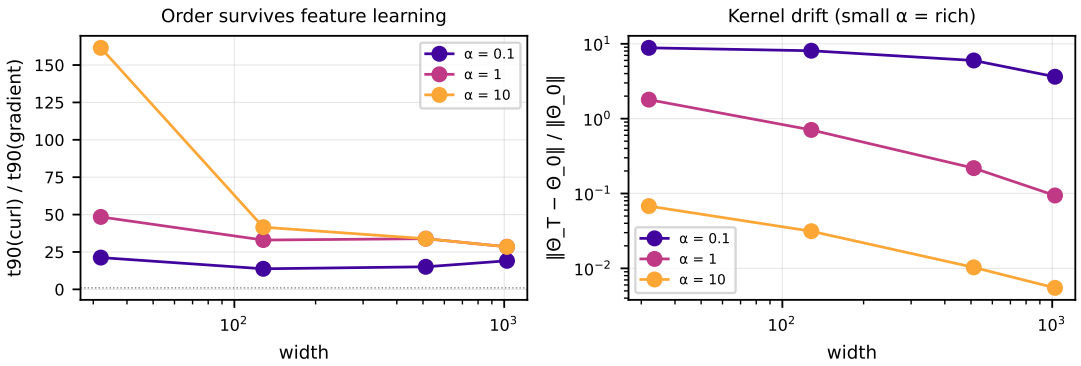


Figure 4: Antisymmetric net across width and output scale α (small α = rich, feature learning). Left: how many times later curl reaches 90% than the gradient part. Right: relative change of the empirical kernel over training.

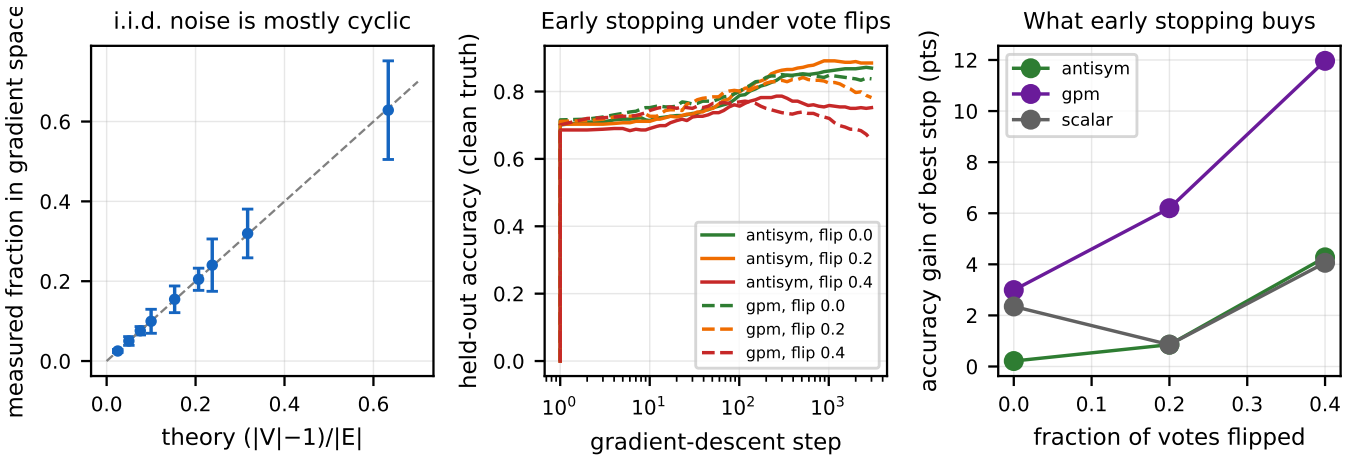


Figure 5: Left: fraction of i.i.d. Gaussian edge noise in gradient space against Lemma 2, over 9 graphs (error bars: one standard deviation over draws). Middle: held-out sign accuracy against the clean truth during training with 0, 20 and 40% of votes flipped (mean of seeds). Right: accuracy gained by stopping at the best step instead of training to the end, by model and flip rate.

and held-out votes are split within pairs, so they test fitting of observed pairs more than generalisation to unseen ones. The NTK stopping rule we proposed failed; a rule that estimates the signal fraction in cycle space, for example with the transitive null of Section 6, is the natural replacement. The full-scale configuration widens every sweep.

Conclusion. Gradient descent learns preferences in a fixed order: the ranking first, the cycles much later. For a scalar reward model the cycles are not merely unrepresentable but invisible to the gradient. For an antisymmetric network the edge NTK puts transitive flows 4 to over 100 times higher in the spectrum than cyclic ones, more so on denser comparison graphs, and training follows that ordering, closely in the lazy regime and robustly beyond it. Local and global cycles, by contrast, are learned at the same rate. The ordering turns training time into a knob on transitivity. It explains why early stopping helps cyclic preference models most when votes are noisy. It also cautions against reading learned cycles as signal: in Chatbot Arena, the cycles a flexible model eventually learns are sampling noise, fitted last.

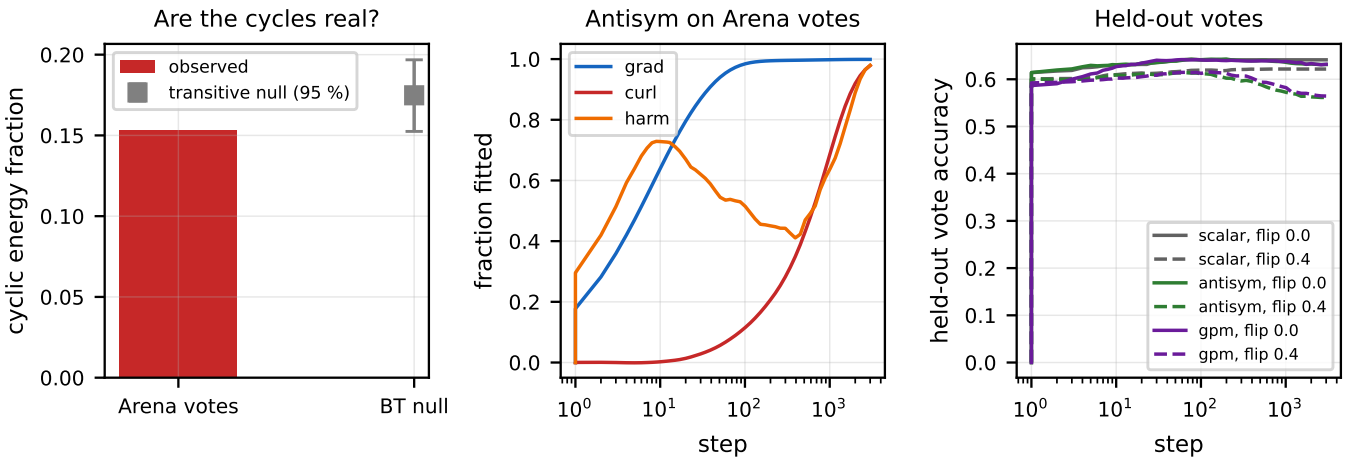


Figure 6: Chatbot Arena, 63 models, 640 model pairs with at least 20 votes. Left: cyclic energy of the observed log-odds flow against a perfectly transitive Bradley–Terry null with the same counts (median and 95% interval). Middle: per-component fit of the antisymmetric net under the Bradley–Terry loss. Right: held-out vote accuracy during training, clean and with 40% of training votes flipped (mean of seeds).

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